

MATH 180 Strategies of Problem Solving

W.R. Casper

Department of Mathematics
California State University Fullerton

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Outline

1 SoPS Lecture 10

- Sprouts
- Graphs
- Wrapping up

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Review

Last time

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- games focused on deductive / spatial reasoning

Sprouts Game

Rules

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- sprouts cannot cross other sprouts
- last player to draw an arc wins

Activity (20 minutes)

Get into groups of two or three and try playing Sprouts!

Observations:

Question:

What kinds of observations can you make about the game Sprouts?

Observations:

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What kinds of observations can you make about the game Sprouts?

Question:

What kinds of conjectures can you make about the game Sprouts?

A single seed:

Question

What happens if the initial setup has only one seed?

Two seeds:

Question

What about if we start with two seeds, what can happen?

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Vocabulary

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Graphs

A **graph** or **network** is a collection of dots called **vertices** and arcs between the dots called **edges**.

Planar graphs

A graph is called **planar** if it can be drawn without the edges crossing each other.

Game length

Question 1

Does every game of Sprouts eventually end? If we start with n seeds, what's the maximum number of moves?

Question 2

When the game has ended, how many regions are enclosed by the final graph?

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Discussion

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Good question

What kind of interesting questions can you think of about Break the Cube?

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What kind of interesting questions can you think of about Break the Cube?

Question

Suppose we wanted to try to count the number of possible arrangements of the three blocks. How might we begin to go about it?

Question of the Day

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Puzzle

Suppose there is an enemy submarine at an unknown location on the integer number line, and each minute, it moves at an unknown, fixed integer velocity. You may fire one missile per minute somewhere on the number line in an attempt to hit the submarine. Is there a strategy to guarantee you will eventually hit the submarine?

Final thoughts

Reflection

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Final thoughts

Reflection

- What was the static pattern you found that worked the best?
- What dynamic algorithm would you suggest to improve your static algorithm.

Submit your responses by **Tonight!**